Math 310 **Fall 2026**
Quiz 4 **Drew Armstrong**

Problem 1. Consider the unit square $D \subseteq \mathbb{R}^2$ defined by $0 \leq x \leq 1$ and $0 \leq y \leq 1$. Compute the center of mass of D if the mass density at the point (x, y) is $\mu(x, y) = x^2$.

The formula for the center of mass is $(\overline{x}, \overline{y}) = (\frac{M_y}{M}, \frac{M_x}{M})$. First we compute the total mass:

$$\begin{aligned} M &= \iint_D \mu(x,y)\, dxdy \\ &= \int_0^1 \int_0^1 x^2\, dx\, dy \\ &= \int_0^1 \left[\frac{x^3}{3}\right]_0^1 dy \\ &= \int_0^1 \frac{1}{3}\, dy \\ &= \frac{1}{3}. \end{aligned}$$

Then we compute the moments about the y-axis and the x-axis:

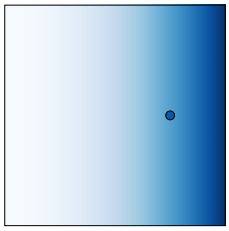
$$\begin{aligned} M_y &= \iint_D x \cdot \mu(x,y)\, dxdy \\ &= \int_0^1 \int_0^1 x^3\, dx\, dy \\ &= \int_0^1 \left[\frac{x^4}{4}\right]_0^1 dy \\ &= \int_0^1 \frac{1}{4}\, dy \\ &= \frac{1}{4}, \end{aligned}$$

$$\begin{aligned} M_x &= \iint_D y \cdot \mu(x,y)\, dxdy \\ &= \int_0^1 \int_0^1 yx^2\, dx\, dy \\ &= \int_0^1 y \left[\frac{x^3}{3}\right]_0^1 dy \\ &= \int_0^1 \frac{y}{3}\, dy \\ &= \frac{1}{6}. \end{aligned}$$

Therefore the center of mass is

$$(\bar{x}, \bar{y}) = \left(\frac{M_y}{M}, \frac{M_x}{M}\right) = \left(\frac{1/4}{1/3}, \frac{1/6}{1/3}\right) = \left(\frac{3}{4}, \frac{1}{2}\right).$$

Here is a picture:



Problem 2. Let $D \subseteq \mathbb{R}^2$ be the right half of the unit disk, defined by $x \geq 0$ and $x^2 + y^2 \leq 1$. Compute the integral

$$\iint_D \sqrt{x^2 + y^2}\, dxdy.$$

We will use polar coordinates with $0 \leq r \leq 1$ and $-\pi/2 \leq \theta\pi/2$. Note that $\sqrt{x^2 + y^2} = r$ and $dxdy = r\, drd\theta$. Hence

$$\begin{aligned}
\iint_D \sqrt{x^2 + y^2}\, dxdy &= \int_{-\pi/2}^{\pi/2} \int_0^1 r \cdot r\, dr\, d\theta \\
&= \int_{-\pi/2}^{\pi/2} \int_0^1 r^2\, dr\, d\theta \\
&= \int_{-\pi/2}^{\pi/2} \left[\frac{r^3}{3}\right]_0^1 d\theta \\
&= \int_{-\pi/2}^{\pi/2} \frac{1}{3}\, d\theta \\
&= \left[\frac{\theta}{3}\right]_{-\pi/2}^{\pi/2} \\
&= \frac{\pi}{3}.
\end{aligned}$$

We can view is as the volume of the solid above the half disk and below the cone $z = \sqrt{x^2 + y^2}$:

